East Swindon Development Group

Land at South Marston

Bats in Trees Survey Report

October 2022

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1.0 INTRODUCTION

1.1 The following bat report has been prepared by FPCR Environment & Design Ltd on behalf of the East Swindon Development Group. It provides details of bat surveys undertaken at land east of Swindon, Wiltshire. Details are provided in relation to measures required to protect the favourable conservation status of bats within the site.

Site Location and Context

1.2 The full Site extends to approximately 142 hectares and lies adjacent to the eastern edge of Swindon. Its southern boundary follows the main Bristol to London Mainline railway, with the A420 further to the south. The northern boundary is formed, in part, by South Marston in its western extent and, in part, by Nightingale Lane and Wood in its eastern extent. The River Cole lies to the east.

1.3 The Study Area covers the western section of the larger site, with the eastern extent following Old Vicarage Lane. This area covers the first phase of development and related haul road and junction works.

2.0 LEGISLATION AND POLICY

2.1 All bats and their roosts are afforded full legal protection under the Conservation of Habitats and Species Regulations 2017 (as amended) and the Wildlife & Countryside Act 1981 (as amended). The purpose of the legislation is to maintain and restore protected species to a situation where their populations are favourable.

2.2 Under Regulation 41 of the Conservation of Habitats and Species Regulations 2017 (as amended) it is illegal to:

- Deliberately capture, injure or kill any wild animal of a European Protected Species (EPS),
- Deliberately disturb wild animals of an EPS (affecting ability to survive, breed or rear young) – disturbance of animals includes in particular any disturbance which is likely to impair their ability to survive, to breed or reproduce, or to rear or nurture their young,
- Deliberately disturb wild animals of an EPS (impairing ability to migrate or hibernate) – disturbance of animals includes in particular any disturbance which is likely to impair their ability in the case of hibernating or migratory species to hibernate or migrate,
- Deliberately disturb wild animals of an EPS (affecting local distribution and abundance) – disturbance of animals includes in particular any disturbance which is likely to affect significantly the local distribution or abundance of the species to which they belong,
- Deliberately disturb wild animals of an EPS (whilst occupying a structure of place used for shelter or protection) – intentionally or recklessly disturb any wild animal while it is occupying a structure or place which it uses for shelter or protection,
- Damage or destroy a breeding site or resting place of a wild animal an EPS.

2.3 In addition, under the Wildlife and Countryside Act 1981 (as amended) it is illegal to:

- Recklessly or intentionally kill, injure or take any wild animals included in Schedule 5.
- Recklessly or intentionally damage or destroy, or obstruct access to any structure or place which any wild animal included in Schedule 5 uses for shelter or protection,

- Recklessly or intentionally disturb any such animal while it is occupying a structure or place which it uses for shelter or protection.

3.0 METHODOLOGY

Tree Assessment

3.1 Trees which were to be impacted by the proposed works were assessed from ground level, with the aid of a torch and binoculars (where appropriate). These surveys were undertaken on 30th September 2021 by a suitably experienced ecologist (Helen Ruffhead, NE Level 1 Bat Licence 2019-43278-CLS-CLS). During the survey Potential Roosting Features (PRF) for bats such as the following were sought (Based on P16, British Standard 8596:2015 Surveying for bats in trees and woodland, October 2015):

- Natural holes (e.g. knot holes) arising from naturally shed branches or branches previously pruned back to a branch collar.
- Man-made holes (e.g. cavities that have developed from flush cuts or cavities created by branches tearing out from parent stems).
- Woodpecker holes.
- Cracks/splits in stems or branches (horizontal and vertical).
- Partially detached, loose or bark plates.
- Cankers (caused by localised bark death) in which cavities have developed.
- Other hollows or cavities, including butt rots.
- Compression of forks with occluded bark, forming potential cavities.
- Crossing stems or branches with suitable roosting space between.
- Ivy stems with diameters in excess of 50mm with suitable roosting space behind (or where roosting space can be seen where a mat of thinner stems has left a gap between the mat and the trunk).
- Bat or bird boxes.
- Other suitable places of rest or shelter.

3.2 Certain factors such as orientation of the feature, its height from the ground, the direct surroundings and its location in respect to other features may enhance or reduce the potential value.

3.3 Trees were classified into general bat roost potential groups based upon the presence of these features. Table 1 (below) broadly classifies the potential categories as accurately as possible as well as discussing the relevance of the features. This table is based upon Table 4.1 and Chapter 6 in Bat Surveys for Professional Ecologists: Good Practice Guidelines (J., Collins (Bat Conservation Trust), 2016).

3.4 Although the British Standard 8596:2015 document groups trees with moderate and high potential, these have been separated below (as per Table 4.1 in The Bat Conservation Trust Guidelines) to allow more specific survey criteria to be applied.

Table 1: Classification and Survey Requirements for Bats in Trees

Classification of Tree	Description of Category and Associated Features (based on Potential Roosting Features listed above)	Likely Further Survey work / Actions
Confirmed Roost	Evidence of roosting bats in the form of live / dead bats, droppings, urine staining, mammalian fur oil staining, etc.	A Natural England derogation licence application will be required if the tree or roost site is affected by the development or proposed arboricultural works. This will require a combination of aerial assessment by roped access bat workers (where possible, health and safety constraints allowing) **and** nocturnal survey during appropriate periods (e.g. nocturnal survey - May to August) to inform on the licence. Works to tree undertaken under supervision in accordance with the approved good practice method statement provided within the licence. **However,** where confirmed roost site(s) are not affected by works, work under a precautionary good practice method statement may be possible.
High Potential	A tree with one or more Potential Roosting Features that are obviously suitable for larger numbers of bats on a more regular basis and potentially for longer periods of time due to their size, shelter protection, conditions (height above ground level, light levels, etc) and surrounding habitat. Examples include (but are not limited to); woodpecker holes, larger cavities, hollow trunks, hazard beams, etc.	Aerial assessment by roped access bat workers (if appropriate) **and** / **or** nocturnal survey during appropriate period (May to August). Following additional assessments, tree may be upgraded or downgraded based on findings. If roost sites are confirmed and the tree or roost is to be affected by proposals a licence from Natural England will be required. After completion of survey work (and the presence of a bat roost is discounted), a precautionary working method statement may still be appropriate.
Moderate Potential	A tree with Potential Roosting Features which could support one or more potential roost sites due to their size, shelter protection, conditions (height above ground level, light levels, etc) and surrounding habitat but unlikely to support a roost of high conservation status (i.e. larger roost,	A combination of aerial assessment by roped access bat workers **and / or** nocturnal survey during appropriate period (May to August). Following additional assessments, tree may be upgraded or downgraded based on findings.

Classification of Tree	Description of Category and Associated Features (based on Potential Roosting Features listed above)	Likely Further Survey work / Actions
	irrespective of wider conservation status). Examples include (but are not limited to); woodpecker holes, rot cavities, branch socket cavities, etc.	After completion of survey work (and the presence of a bat roost is discounted), a precautionary working method statement may still be appropriate. If a roost site/s is confirmed a licence from Natural England will be required.
Low Potential	A tree of sufficient size and age to contain Potential Roosting Features but with none seen from ground or features seen only very limited potential. Examples include (but are not limited to); loose/lifted bark, shallow splits exposed to elements or upward facing holes.	No further survey required but a precautionary working method statement may be appropriate.
Negligible/No potential	Negligible/no habitat features likely to be used by roosting bats	None.

* The Conservation of Habitats & Species Regulations 2017 (as amended) affords protection to "breeding sites" and "resting places" of bats. The EU Commission's Guidance document on the strict protection of animal species of Community interest under the Habitats Directive 92/43/EEC, 2021 states that these are places "where there is a reasonably high probability that the species concerned will return".

Aerial Tree Inspections

3.5 Trees identified during the ground level inspection as providing features suitable to be used as a bat roost were subject aerial inspections where required and safe to do so. These inspections were conducted on the 1st and 2nd February 2022 by Sam Arthur (Natural England Level 2 bat licence 2015-19188- CLS) and an assistant (James Hutchison). Each feature suitable for roosting bats was visually inspected using torches and/or endoscopes as appropriate. The characteristic of each feature was considered to assess its suitability to support roosting bats. The size and exposure to the elements of each was additionally taken into account, as were features such as dense cobwebs or the habitation of a feature by other species (e.g. woodpeckers, squirrels, wasps etc.).

3.6 All FPCR tree climbers are NPTC Certified to Climb Trees (J/101/2449) and Perform Aerial Rescue (A/101/2450)–Level 2. The climbing methodology used follows that detailed within the Arboriculture and Forestry Advisory Group (AFAG) Tree Climbing Operations Leaflet (AFAG401). Climbing equipment was inspected following guidelines outlined in the Lifting Operations and Lifting Equipment Regulations 1998.

Nocturnal Surveys

3.7 Nocturnal dusk emergence and dawn re-entry surveys were completed by suitably experienced ecologists from FPCR between 9th May 2022 and 1st August 2022. Surveyors were positioned at various aspects of the trees from approximately 15 minutes prior to sunset to 90 minutes after sunset, or from 90 minutes prior to sunrise to 15 minutes after. The number and species of bats observed emerging or entering the tree was recorded.

3.8 Wildlife Acoustics Inc. Echo Meter Touch® bat detectors in conjunction with the Echo Meter Touch® app were used by surveyors to aid in identification. All of the nocturnal surveys were conducted in appropriate conditions, i.e. ambient temperature exceeding 10˚C and little wind/rain.

Additional Surveys

3.9 An additional endoscope inspection survey was carried out on potential roost features on tree TG28b on the 1st August 2022. This was conducted by Anastasia Spencer (Natural England Level 4 bat licence 2021-10045-CL20-BAT).

4.0 RESULTS

Tree Assessment

4.1 The trees on site were a mix of young to mature specimens and of varying structural state (It should be noted that ecological and arboricultural survey and assessment methodologies and aims differ).

4.2 Fifteen trees were identified during the ground assessment as having potential for roosting bats. These were subject to aerial inspections where possible, and a final roost potential assessment was made. Details of these features are in Table 2 below, with locations in Figure 1.

Table 2: Trees with Bat Potential

Tree ref.	Species	Potential bat roost features (distance above ground and aspect where appropriate)	Final Bat Roost Potential Category (as prescribed in Table 1)	Further Work
TG28a	Crack Willow, Salix fragilis	**E:** Rot hole at 2m. 50cmx20cm entrance leads to large cavity approx. 30cm diameter and 150cm tall.	High	Nocturnal surveys.
TG28b	Crack Willow, Salix fragilis	**W:** Rotten stump with vertical split to 2m. Multiple small cracks internally.	High	Nocturnal surveys.
T35	English Oak, Quercus robur	**S:** Tear out wound at 4m **S:** Woodpecker hole at 3m **S:** Knot hole at 10m	High	Nocturnal surveys.
T50	Crack Willow, Salix fragilis	**N:** Hollow trunk with vertical split to 3m opening to multiple smaller cavities. **S:** Woodpecker hole at 7m. 5cm diameter entrance extending back and up for 15cm.	High	Nocturnal surveys.

Tree ref.	Species	Potential bat roost features (distance above ground and aspect where appropriate)	Final Bat Roost Potential Category (as prescribed in Table 1)	Further Work
T54	English Oak, Quercus robur	**E:** Hazard beam at 5m. **W:** Open hollow branch end at 10m **W:** Platey lifted bark at 4m.	High	Nocturnal surveys.
T23	Ash, Fraxinus excelsior	**W:** Knot hole at 14m in dead limb.	Moderate	Nocturnal surveys.
T34	English Oak, Quercus robur	**SW:** Knot hole at 3m. 3cm entrance diameter extending 15cm horizontally then upwards 20cm.	Moderate	Nocturnal surveys.
T44	Ash, Fraxinus excelsior	**E:** Woodpecker hole at 3m **W:** Butt rott at 3m opening downwards to a large internal cavity. Partially exposed but sheltered sections.	Moderate	Nocturnal surveys.
T49	Crack Willow, Salix fragilis	**S:** Cavity at base of main trunk reaching 30cm. Internal dimensions 30cm high by 5cm wide.	Moderate	Nocturnal surveys.
T55	English Oak, Quercus robur	**S:** Horizontal crack at 6m. Cavities present at both ends	Moderate	Nocturnal surveys.
T108	Ash, Fraxinus excelsior	**SW:** Branch tear out at 4m. 5cm diameter entrance cavity. **E:** Vertical crack at 3m along underside of limb.	Moderate	Nocturnal surveys.
T2	Crack Willow, Salix fragilis	Failed pollard with splits through trunk base from ground level.	Low	No further survey work. Precautionary felling method
TG3a	Apple, Malus domestic a	**W:** Woodpecker hole at 3m. 5cm diameter extends 10cm back and 20cm down with water pooling at the base.	Low	No further survey work. Precautionary felling method
T56	English Oak, Quercus robur	**W:** Dead wood at 4m. Small cavities apparent.	Low	No further survey work. Precautionary felling method
T107	Ash, Fraxinus excelsior	**S:** Small rot hole at 2.5m in branch.	Low	No further survey work. Precautionary felling method

Nocturnal survey

4.3 Eleven trees were identified to provide moderate or high potential for roosting bats. These trees were subject to nocturnal survey as detailed in Table 3 below.

Table 3: Nocturnal Survey Details (2022)

Tree ref	Survey Details	Bat Roost Present	Activity Details
TG28a	10th May (Dusk) 25th May (Dusk) 7th June (Dawn)	No	
TG28b	25th May (Dusk) *7th June (Dawn)* 5th July (Dawn) 1st August (Dusk)	*No*	Possible bat reported on 7th June. Additional survey effort undertaken to discount likely roost. Precautionary felling method.
T35	**11th May (Dusk)** 26th May (Dawn) **6th June (Dusk)**	**YES**	Single noctule roosting in woodpecker hole at 10m. Bat confirmed on two survey occasions.
T50	9th May (Dusk) 25th May (Dawn) 5th July (Dawn)	No	
T54	10th May (Dawn) 24th May (Dusk) 7th June (Dusk)	No	
T23	11th May (Dusk) 25th May (Dusk)	No	
T34	17th May (Dawn) 7th June (Dusk)	No	
T44	12th May (Dawn) 26th May (Dusk)	No	
T49	9th May (Dusk) 25th May (Dawn)	No	
T55	10th May (Dawn) 24th May (Dusk)	No	
T108	12th May (Dawn) 26th May (Dawn)	No	

5.0 DISCUSSION

Tree T35

5.1 Tree T35 is an English oak *Quercus robur* which sits adjacent to the haul road route. The tree has several south facing rot holes and a split along a dead limb providing moderate bat roost potential.

Three nocturnal surveys were conducted. A single noctule bat *Nyctalus noctule* was recorded using a south facing knot hole cavity at approximately 10m, on two of the three survey occasions.

5.2 While the tree is to be retained, the increase in site activity has the potential for causing disturbance to roosting bats, which may in turn lead to roost abandonment. The impact of abandonment would not cause a significant impact to the favourable conservation status of noctule bats in the local area, nor would direct harm be inflicted to the bat itself.

5.1 To mitigate the risk of potential loss of suitable roosting habitat, three bat boxes (2FN Schwegler or equivalent) should be installed on retained trees in the habitat enhancement area prior to works commencing (Figure 2). Boxes should be mounted at a minimum height of 3m with a clear flightpath from the boxes, and lighting schemes should ensure these boxes are not lit to encourage use (as detailed in section 5.10 and 5.11 below).

5.2 Boxes must be installed by / under the supervision of an experienced or licensed bat ecologist to ensure they are installed correctly and in optimum locations.

5.3 A root protection buffer should be maintained around tree T35 to avoid accidental damage.

Tree TG28b

5.4 Tree TG28b is part of a group of eight crack willow *Salix fragilis* trees, recorded as G28 in the Tree Protection Plan (FPCR 2022). It is an old pollard with a vertical split running up the west side and numerous small cracks within the rotting stump.

5.5 The tree was initially classified to provide moderate roosting potential and subject to two nocturnal surveys. An unidentified bat was seen approaching the tree on the second survey occasion and was suspected of entering the vertical split. No audio signal was detected during this time. As a result, an enhanced survey effort was undertaken, totalling four nocturnal surveys and an additional endoscope inspection of all cavities. No further evidence of bat use was recorded (neither observed bat roosting behaviour, or physical evidence such as droppings, scratch marks, smoothing or staining).

5.6 The European Commission's Guidance document on the strict protection of animal species of Community interest under the Habitats Directive (2021) states in paragraph 54 that breeding sites and resting places need to be protected where "*there is a reasonably high probability that the species concerned will return to these sites and places*". The example used to illustrate this point being a cave roost used for winter hibernation.

5.7 Based on the total survey effort and strength of observations, it was concluded that it is unlikely that the observed bat will return to tree TG28b.

5.8 As a proportionate precaution, a pre-works check by a qualified bat worker should be undertaken to examine the potential roost features immediately prior to the tree's removal.

5.9 If no evidence of bat use is discovered, felling work should proceed following a sectional felling method. Arisings should be removed from the working area and used in the adjacent retained wooded habitat enhancement area to create deadwood habitat resources such as log piles and individual grounded large wood trunks / branches (figure 2). The main stem in which the potential roost features are located should be left in situ after felling for 24 hours before being moved to the habitat enhancement area.

Artificial Lighting

5.10 Works during the active bat season (April – October inclusive) adjacent to retained linear edge habitats on or off-site should be carried out during daylight hours wherever possible, however where this is not possible due to particular programming requirements, lighting should be directional and avoid light spill onto these habitats to prevent adversely affecting foraging and commuting habitats utilised by bats. All lighting must be turned off when construction Site is not active.

5.11 The design of lighting is based upon guidelines presented in the Institute of Lighting Professionals *'Guidance Note 8 Bats and Artificial Lighting'* 2018, Bat Conservation Trust & Institute of Lighting Engineers *'Bats and Lighting in the UK - Bats and Built Environment Series',* the Bat Conservation Trust *'Artificial Lighting and Wildlife Interim Guidance'* and the Bat Conservation Trust *'Statement on the impact and design of artificial light on bats'*. Therefore, the lighting scheme will include the following:

- The strategic use of landscaping and planting to avoid light spill on sensitive habitats. All edges of the site boundary should be buffered with additional planting to prevent light spill which will encourage use by bats. Additional LUX levels should be no greater than 1LUX for this habitat area.
- The avoidance of direct lighting of existing trees, scrub, woodland, or proposed areas of habitat creation / landscape planting. This is particularly important for this site due to the presence of light sensitive barbastelle bats. Any lighting should be directed away from retained or created habitats.
- Unnecessary light spill will be controlled through a combination of directional lighting, low level lighting columns, hooded / shielded luminaires and strategic planting. Smart lighting should be considered for footpaths and areas of lighting around the edges of the development area to ensure lighting is only provided when necessary.
- All new column mounted luminaires shall be fitted with flat glass where appropriate to aid 0% upward light discharge which will reduce light pollution for larger bat species found foraging over the site such as noctule and serotine.
- Where located adjacent to green corridors and retained habitats, lights should be installed facing away from the habitats and will be fitted with rear light baffles or cowls to prevent light spill behind them.

6.0 CONCLUSION

6.1 Fifteen trees impacted by the proposals were assessed to provide a degree of bat roost potential. Bat activity was recorded in association with two of these trees: T35 with a confirmed roost of a single noctule, and TG28b with a single bat sighting recorded potentially entering the tree, but later confirmed as having no active roost with the bat unlikely to return.

6.2 In following the advice provided in this report, including tree retention; the use of a sensitive-lighting scheme; pre-work checks and soft-felling, and installation of suitable bat boxes, it is expected that effects of development on the local bat populations can be sufficiently mitigated, with the additional creation of log piles in the retained enhancement area providing extra foraging resources for bats by way of suitable habitat for their prey (invertebrates).

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Key

Site Boundary

Trees with bat roost potential (based on aerial assessment)

- High
- Moderate
- Low
- See report for details on bat activity

0 50 100 150 200 m

client
East Swindon Development Group

project
Land at South Marston

drawing title
Bat Roost Potential Assessment

scale @ A3
1:3,800

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Figure 1

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masterplanning ■ environmental assessment ■ landscape design ■ urban design ■ ecology ■ architecture ■ arboriculture

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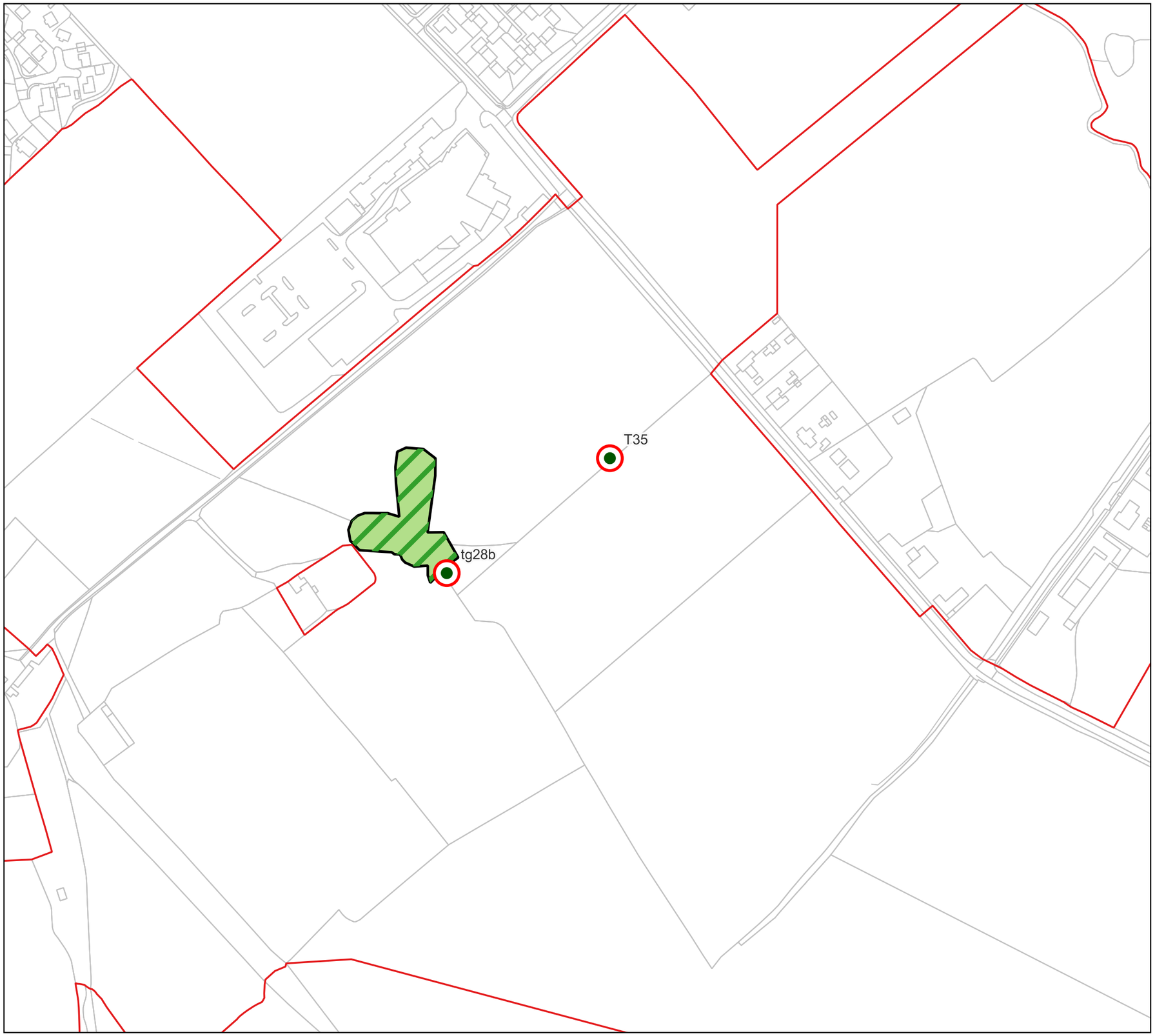

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Key

- Trees with Bat Constraints
- Habitat Enhancement Area

client
East Swindon Development Group

project
Eastern Villages, Swindon

drawing title
Habitat Enhancement Plan

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Figure 2

N

0 50 100 150 200 m

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